Lecture 24
Decisions; tests for p

Review

1 **A drug maker says 80% of patients improve. A doctor suspects fewer do. State H_0 and H_A.**

H_0: $p = 0.80$ H_A: $p < 0.80$ ($p_0 = 0.80$, lower-tailed)

2 **What is the test statistic z_{obs}?**

How many SEs the estimate is from θ_0: $z_{obs} = (\text{estimate} - \theta_0) / \text{SE}$

3 **What is the P-value?**

The probability of a result as or more extreme than z_{obs}, if H_0 is true

Making a decision

significance level α

the cutoff for "rare enough", chosen before collecting data; usually $\alpha = 0.05$

P-value $< \alpha$

reject H_0: the result is statistically significant, evidence for H_A

P-value $\geq \alpha$

do not reject H_0: the data are not convincing evidence for H_A

"do not reject" is not "accept"

not rejecting H_0 does not prove it true; the evidence just isn't strong enough

Practice: P-values and decisions

For each test, write the P-value as a probability given H_0 is true. Then use its value to make a decision.

P-value = P(? | H_0 true)

(a)	**H_0: p = 0.40**	**H_A: p > 0.40**	**z_{obs} = 1.52**	**α = 0.05**	**P-value = 0.0643**
(b)	**H_0: μ = 12**	**H_A: μ < 12**	**t_{obs} = −2.30, df = 15**	**α = 0.01**	**P-value = 0.0181**
(c)	**H_0: μ = 50**	**H_A: μ ≠ 50**	**z_{obs} = −2.10**	**α = 0.05**	**P-value = 0.0358**

(a) P(Z ≥ 1.52 | H_0 true) = 0.0643 ≥ 0.05: do not reject H_0
(b) P(T ≤ −2.30 | H_0 true) = 0.0181 ≥ 0.01: do not reject H_0
(c) P(Z ≤ −2.10 or Z ≥ 2.10 | H_0 true) = 2(0.0179) = 0.0358 < 0.05: reject H_0

Testing a population proportion

when to use it

a categorical variable, and a claim H_0: $p = p_0$ about the proportion of successes

assumptions

a random sample, and $np_0 \geq 15$ and $n(1 - p_0) \geq 15$

test statistic

$z_{obs} = (\hat{p} - p_0) / \sqrt{(p_0(1 - p_0) / n)}$: the SE uses p_0, because we assume H_0 is true

P-value

upper: $P(Z \geq z_{obs} \mid H_0 \text{ true})$; lower: $P(Z \leq z_{obs} \mid H_0 \text{ true})$; two-sided: double the tail

Example: cell phones and brain cancer

In a study of 420,019 cell phone users, 172 developed brain cancer. For people who do not use cell phones, the rate is 0.034%. Do cell phone users get brain cancer at a higher rate? (OpenStax, Example 9.21)

$x = 172$; $n = 420{,}019$; $p_0 = 0.00034$; $\alpha = 0.05$

(a) What are H_0 and H_A? Which tail? Are the assumptions met?
(b) What are $\hat{p}$, the standard error and z_{obs}?
(c) What is the P-value? What is the decision at $\alpha = 0.05$?
(d) The book uses $\alpha = 0.005$. Would the decision change?

(a) $H_0: p = 0.00034$ $H_A: p > 0.00034$, upper-tailed; $np_0 = 142.8 \geq 15$, $n(1 - p_0) \geq 15$
(b) $\hat{p} = 172 / 420{,}019 \approx 0.0004095$; $SE = \sqrt{p_0(1 - p_0) / n} \approx 0.00002845$
$z_{obs} = (0.0004095 - 0.00034) / 0.00002845 \approx 2.44$
(c) $P(Z \geq 2.44 \mid H_0 \text{ true}) = 1 - 0.9927 = 0.0073 < 0.05$: reject H_0
(d) Yes: $0.0073 \geq 0.005$, do not reject H_0. Same data, opposite decision: choose α first